Wi-Fi network meter Pico W and XIAO ESP32-C5

Shows how crowded each 2.4 GHz Wi-Fi channel is, so you can pick the best channel for your router.

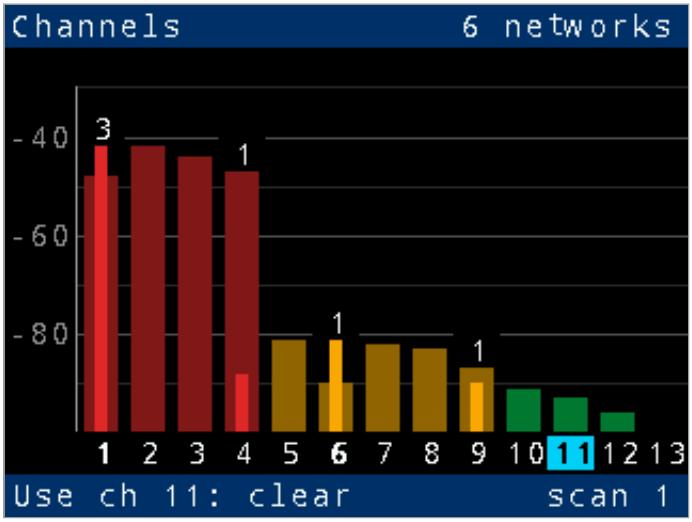
Buttons

A	Next screen
B	Previous screen
X	Channels: scan now. Networks, Events: previous page
Y	Networks, Events: next page

LED

Blue	Scanning
Green	Recommended channel is clear
Amber	Recommended channel is moderately busy
Red	Even the best channel is busy
Magenta	Last scan failed (retried in 2 s; restarts after 5 in a row)

Channels screen



- **Bright bar**: signal, the strongest network on the channel.
- **Dim bar**: interference, every other network on the channel or up to 3 channels away.
- **Scale**: dBm, with a grid line every 10 dB.
- **Colour**: how busy, counting both: green clear, amber moderate, red busy.
- **Number above a bar**: networks on that channel.
- **Bold channel numbers**: the standard channels 1, 6 and 11, which don't overlap.
- **Highlighted channel**: the one to use, the quietest of 1, 6 and 11. The footer names it: "Use ch 11: clear".

Networks screen



- Strongest first: **name**, **channel**, **signal** in dBm (closer to 0 is stronger).
- **Signal colour**: green good (≥ −60), amber fair (≥ −75), red weak.
- Names over 24 characters are shortened in the middle, keeping the end.
- 9 to a page: Y next page, X previous; the footer shows which ("10-13 of 13").
- Names in **amber** are open networks (no password).
- Names in **grey** are hidden networks, shown by their MAC address: "hidden 02:1f:3c:a2:7f:06".

Events screen

Each start, newest first, with the run time before it: **power on** (white); **Wi-Fi restart** (amber); **crash**, **scan/main hung** or **restart** with no reason (red). XIAO also: **HW watchdog**, **CPU fault**, **brown-out**. Kept in flash. X+Y: whole log to the serial console.

Fading

A network stays at full brightness for **60 s** after it was last heard, then fades to 40% and is gone at **2 min**. Dim bars and rows are old data. A network that is heard again is bright again straight away.

Good to know

- Scans run every 10 s (X on Channels: now). Measure where you use Wi-Fi, not by the router.
- 2.4 GHz only: the Pico W cannot hear 5 GHz networks; the XIAO shows 2.4 GHz for now.
- Keep it away from USB 3 hubs: their noise drowns out all but the strongest networks. Use a long cable or a USB charger.
- Serial log on the board's USB socket: `screen /dev/cu.usbmodem<N> 115200` on a Mac.